

Tolga Bilgis

Linux Systems | Infrastructure | HPC

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EDUCATION

Clemson University

Expected May 2028

B.S. Computer Engineering | Minor in Mathematical Sciences | GPA: 3.9/4.0

TECHNICAL SKILLS

Systems & networking: Linux, systemd, TCP/IP, DNS, HTTP/TLS, SSH, NFS

Automation & delivery: Ansible, Terraform, cloud-init, Git, GitHub Actions, GitLab CI/CD, Argo CD

Cloud & containers: AWS (EC2, VPC, IAM, S3), Kubernetes, Docker, Podman, Helm

Observability: Prometheus, Grafana, Alertmanager, OpenTelemetry, Loki

HPC: Slurm, MPI, OpenMP, InfiniBand/RDMA, Spack

Languages: Python, Bash, C, C++, Go, SQL, JavaScript

EXPERIENCE

HPC Intern - AI Infrastructure

May - Aug 2026

Los Alamos National Laboratory | Los Alamos, NM

- Built a 10-node stateless HPC cluster with Slurm; automated provisioning with cloud-init and Ansible and boot-image builds with GitLab CI/CD.
- Built a retrieval-augmented generation (RAG) system for 2 TB of scientific PDFs; benchmarked five vector databases on 2.7 million 4,096-dimensional embeddings.
- Selected and integrated a retrieval backend delivering 20x the prior solution's throughput in the evaluated workload.
- Research poster accepted through peer review for presentation at the SC26 Supercomputing Conference.

HPC System Administrator

Dec 2024 - Present

Clemson University - High-Performance Cluster Computing Creative Inquiry | Clemson, SC

- Co-built and administer the student HPC team's cluster with eight NVIDIA A100 GPUs; configure distributed workloads and tune performance using Slurm and MPI.
- Evaluated distributed graph algorithms across three nodes, analyzed scaling and communication bottlenecks, and co-authored the SC25 reproducibility paper.
- Competed in the SC25 Student Cluster Competition; returning for SC26. Built a 16-node Raspberry Pi cluster for the Single Board Cluster Competition.

AI Coding / Math Expert

Jan 2024 - Jan 2026

Outlier AI | Remote, Contract

- Evaluated code and mathematical solutions for correctness; wrote and tested reference solutions in C++, Python, Java, Kotlin, and JavaScript.

Undergraduate Robotics Researcher

Aug - Dec 2024

Clemson University | Clemson, SC

- Integrated robotic vehicle sensors in Python and C++ and built a real-time telemetry dashboard for speed, position, and sensor readings to support a digital twin.

PROJECTS

Linux & Kubernetes Homelab

Jul 2026 - Present

Kubernetes, Ansible, GitHub Actions, Argo CD

- Built a three-node Kubernetes cluster with highly available control-plane services and replicated storage; automated Linux host configuration with Ansible.
- Host my portfolio on the cluster with a CI/CD pipeline: image tests, GHCR publishing, digest-pinned deployment PRs, and Argo CD reconciliation; expose HTTPS through Cloudflare Tunnel.
- Deployed Prometheus, Grafana, and alerts for service health and resource usage; tested node failures and backup restores, verifying data integrity.

Distributed Edge AI Theft Detection

Feb 2026

NVIDIA Jetson, Docker, YOLO

- Built a distributed theft detection system on two NVIDIA Jetsons in 24 hours, splitting object detection and facial recognition across nodes and displaying alerts in a web dashboard.
- Containerized GPU inference with Docker and a reusable launch script; earned 2nd Place, Best Use of Hardware at CUHackit 2026.